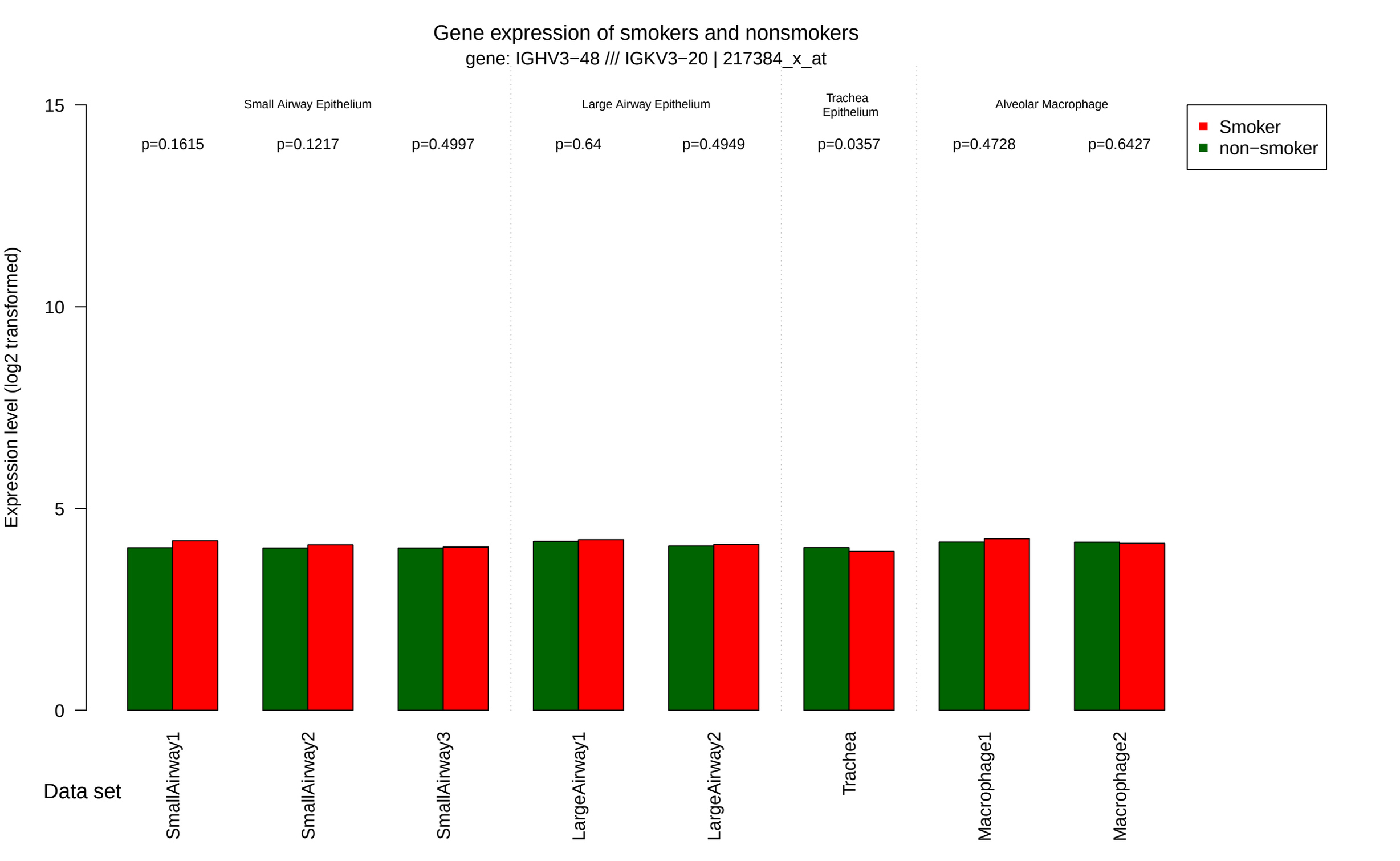
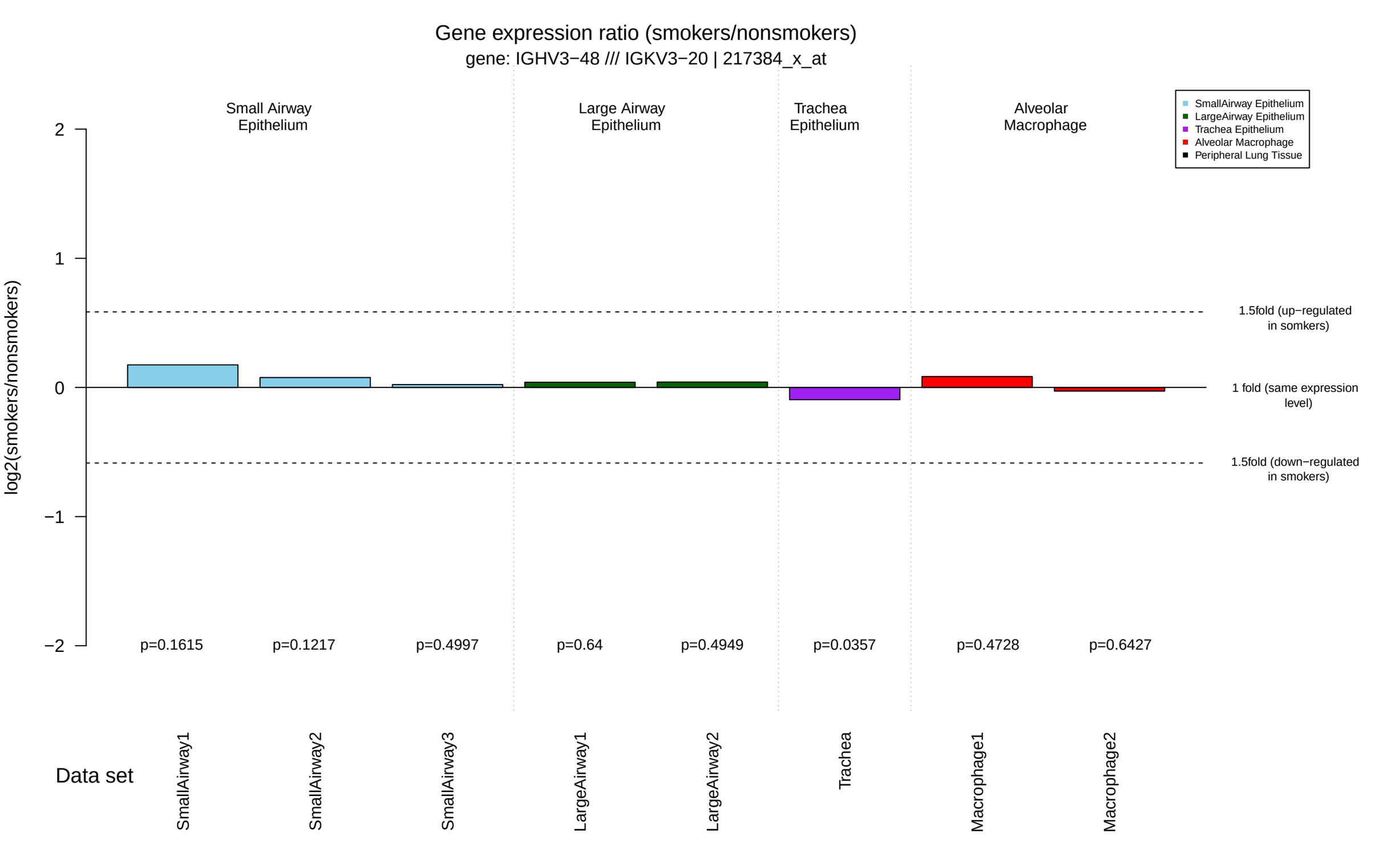
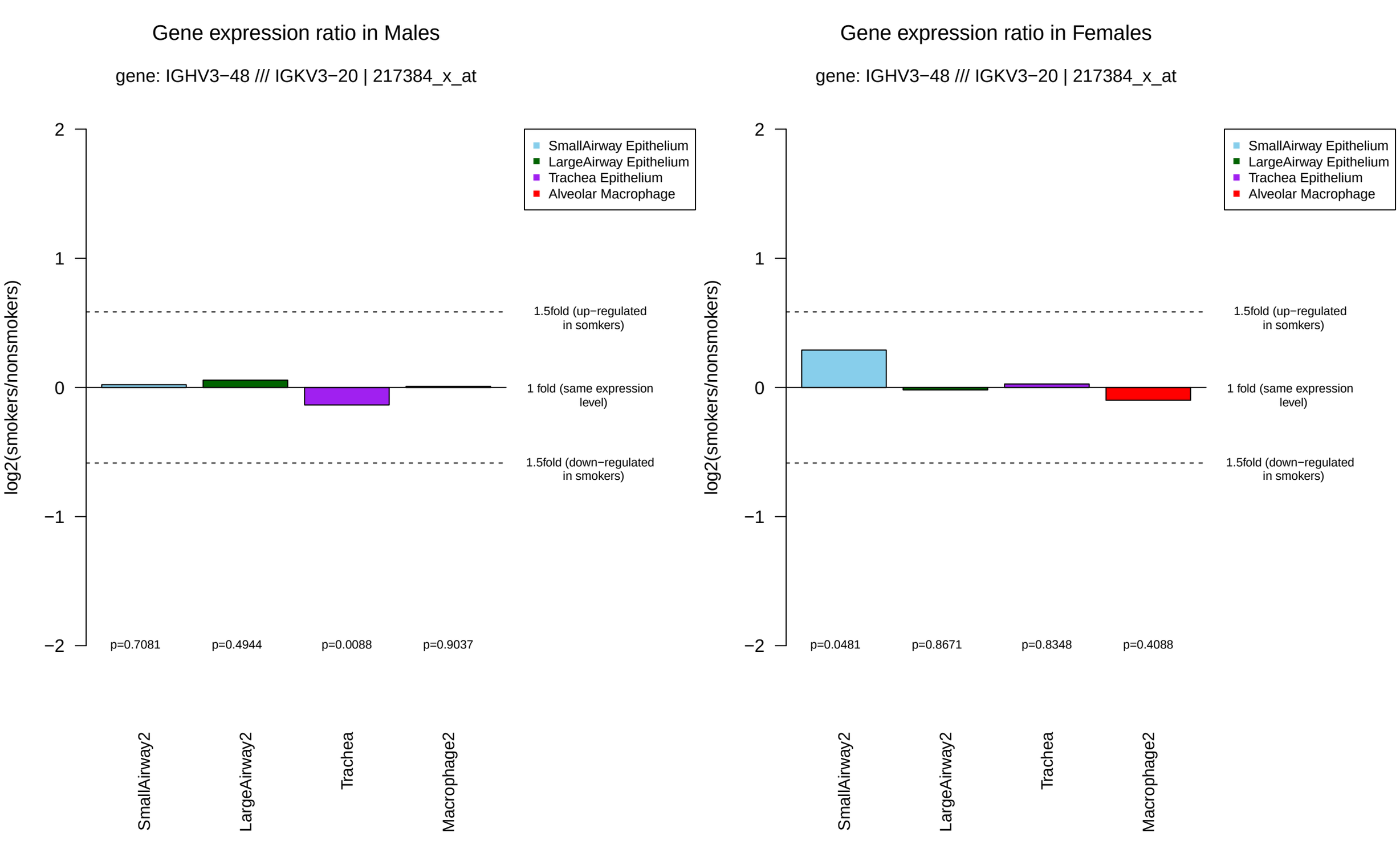
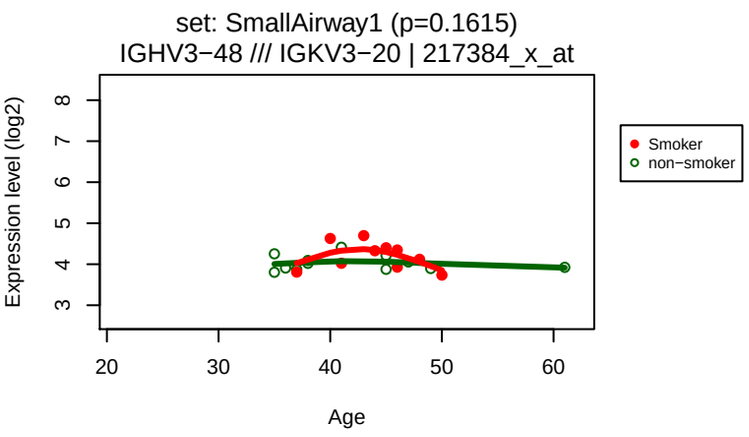
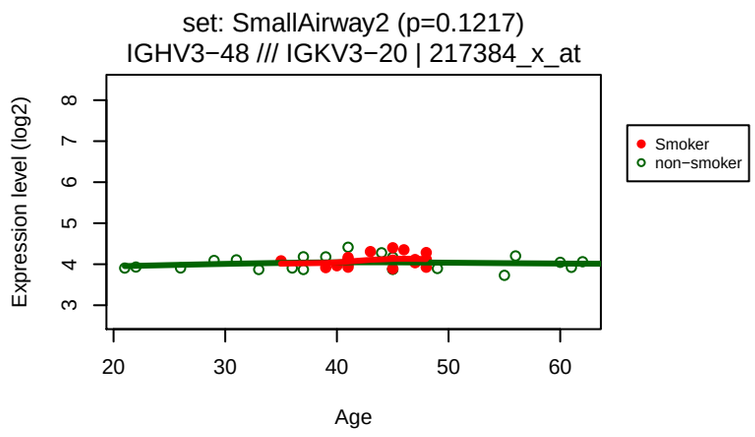
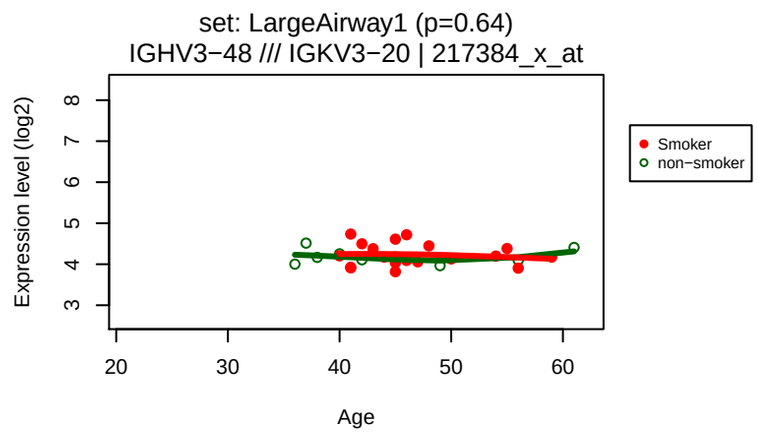

Gene expression of smokers and nonsmokers
gene: IGHV3−48 /// IGKV3−20 | 217384_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1615
p=0.1217
p=0.4997
p=0.64
p=0.4949
p=0.0357
p=0.4728
p=0.6427
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

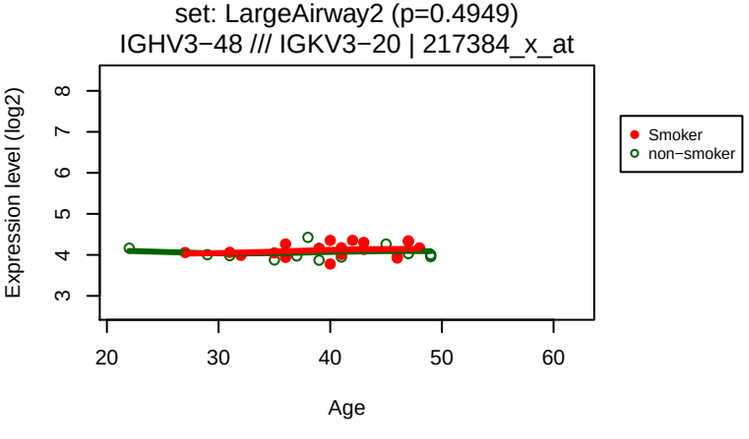

Gene expression ratio (smokers/nonsmokers)
gene: IGHV3−48 /// IGKV3−20 | 217384_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1615
p=0.1217
p=0.4997
p=0.64
p=0.4949
p=0.0357
p=0.4728
p=0.6427
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

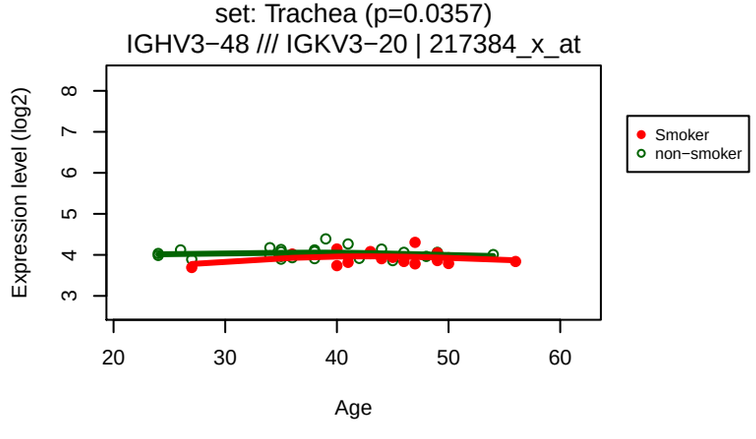

Gene expression ratio in Males
gene: IGHV3−48 /// IGKV3−20 | 217384_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7081
p=0.4944
p=0.0088
p=0.9037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHV3−48 /// IGKV3−20 | 217384_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0481
p=0.8671
p=0.8348
p=0.4088
SmallAirway2
LargeAirway2
Trachea
Macrophage2

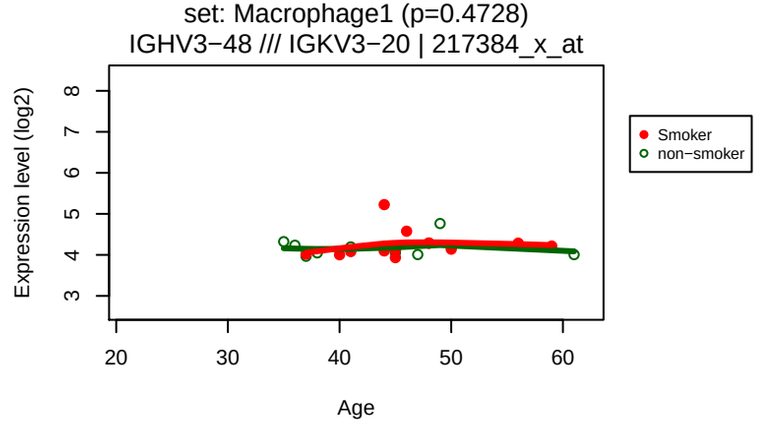

set: SmallAirway1 (p=0.1615)
IGHV3−48 /// IGKV3−20 | 217384_x_at
Expression level (log2)
Age
Smoker
non−smoker

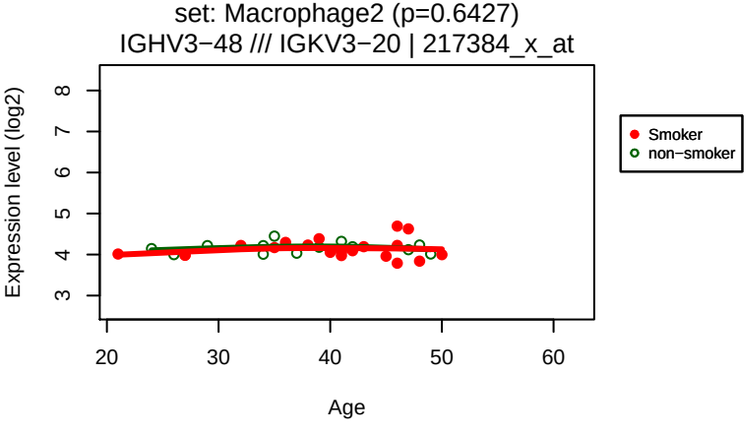

set: SmallAirway2 (p=0.1217)
IGHV3−48 /// IGKV3−20 | 217384_x_at
Expression level (log2)
Age
Smoker
non−smoker

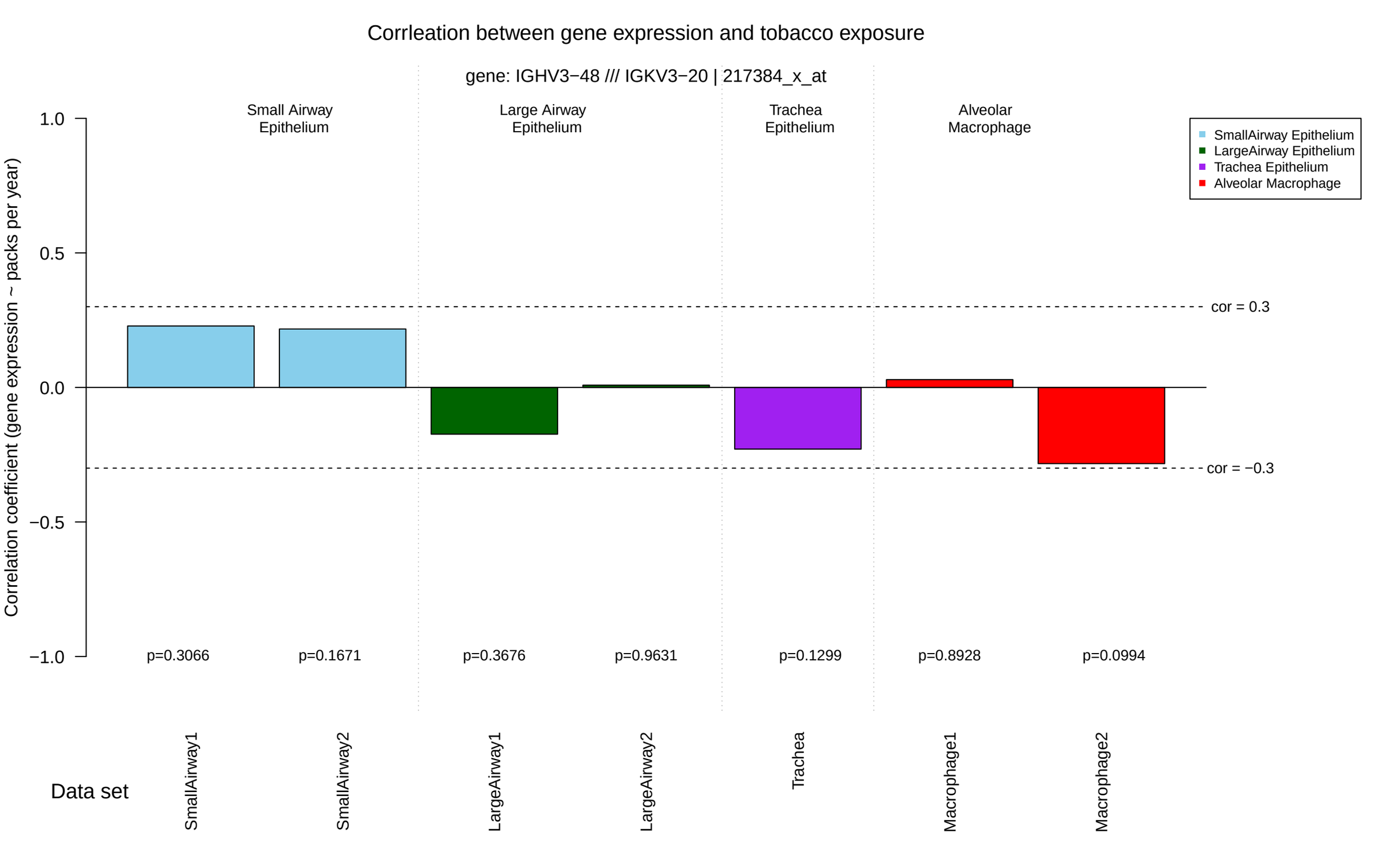

set: LargeAirway1 (p=0.64)
IGHV3−48 /// IGKV3−20 | 217384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4949)
IGHV3−48 /// IGKV3−20 | 217384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0357)
IGHV3−48 /// IGKV3−20 | 217384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4728)
IGHV3−48 /// IGKV3−20 | 217384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6427)
IGHV3−48 /// IGKV3−20 | 217384_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHV3−48 /// IGKV3−20 | 217384_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3066
p=0.1671
p=0.3676
p=0.9631
p=0.1299
p=0.8928
p=0.0994
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2